MAX PLANCK INSTITUTE
FOR DYNAMICS OF COMPLEX
TECHNICAL SYSTEMS
MAGDEBURG

DMP
DATA-DRIVEN MODELING OF
COMPLEX PHYSICAL SYSTEMS

Kinetically Consistent Coarse Graining using Kernel-based Extended Dynamic Mode Decomposition

Feliks Nüske

June 03, 2025

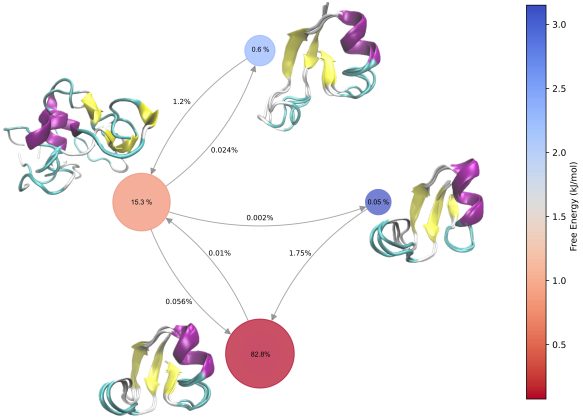
Outline

1. Metastability and Koopman Operators
2. Extracting Kinetics from Data
3. Kernels and Random Features
4. Applications

joint work with: Cecilia Clementi, Weilong Chen, Boumediene Hamzi, Stefan Klus, Selma Moqvist, Vahid Nateghi, Frank Noé, Simon Olsson, Sebastian Peitz, Christof Schütte, Hao Wu

Metastability

- **Metastability** ⇔ systems with **long-lived states** under time evolution:

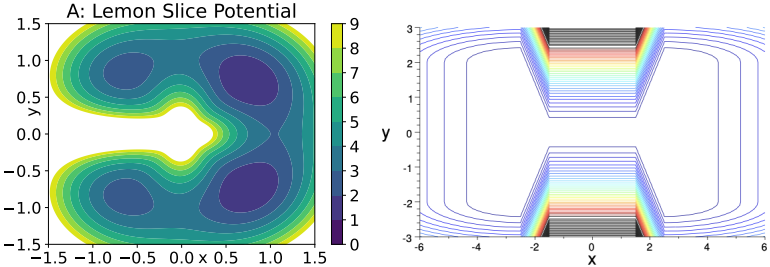


Metastability

- Model system: **Langevin SDE**

$$dX_t = -\nabla V(X_t)\,\mathrm{d}t + \sqrt{2\,k_B T}\,\mathrm{d}W_t.$$

- **Energetic** vs **entropic** origin.

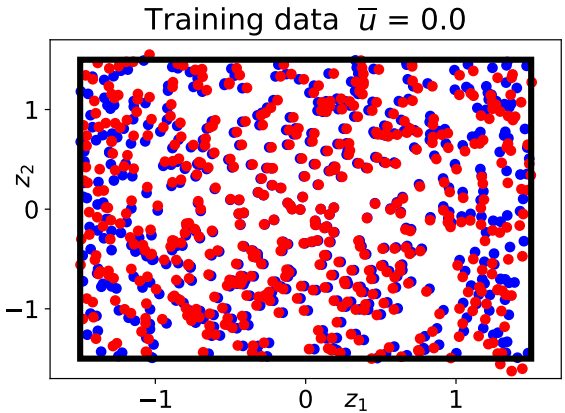


Right figure from **Lelievre & Stoltz**, Acta Numerica, 2016

Koopman Operator

- **Koopman operator** (transfer operator):

$$\mathcal{K}^t \phi(x) = \mathbb{E}\left[\phi(X_t)\,|X_0 = x\right].$$



Koopman, P.N.A.S. (1931), **Schütte et al**, J. Comp. Phys. (1999);
Mezic, Nonlinear Dyn. (2005)

Koopman Operator

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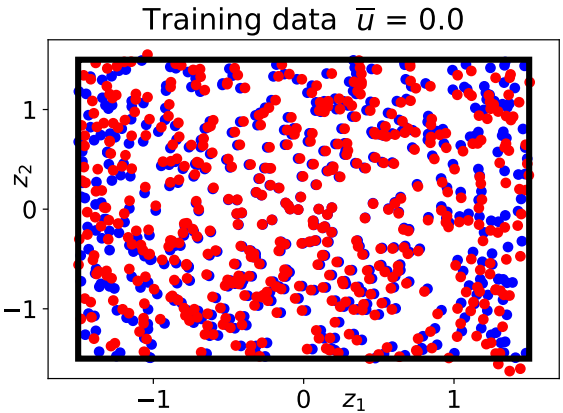
$$\mathcal{K}^t \phi(x) = \mathbb{E}\left[\phi(X_t)\,|X_0 = x\right].$$

- infinitesimal generator:

$$\mathcal{L}\phi = \lim_{h\to 0} \frac{1}{h}(\mathcal{K}^t - \mathrm{Id})\phi.$$

- for Langevin SDE:

$$\mathcal{L}\phi = -\nabla V \cdot \nabla\phi + k_B T \Delta\phi.$$

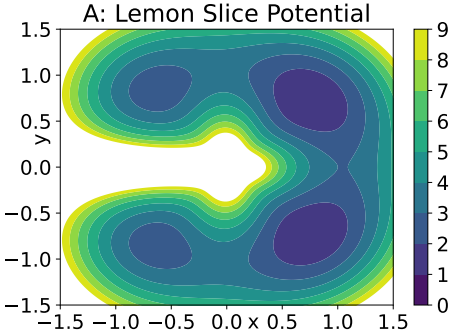


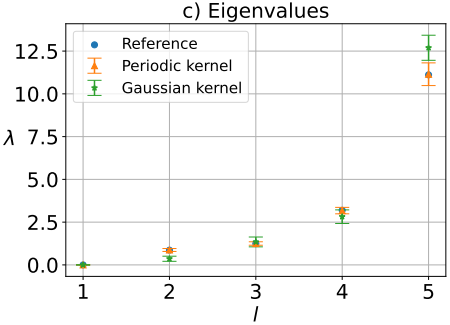
Koopman, P.N.A.S. (1931), **Schütte et al**, J. Comp. Phys. (1999); **Mezic**, Nonlinear Dyn. (2005)

Spectral Analysis

- Metastability: **dominant eigenpairs** of the Koopman operator / generator:

$$\mathcal{L}\psi_i = -\kappa_i \psi_i, \qquad \mathcal{K}^t \psi_i = e^{-\kappa_i t}\psi_t.$$





Davies, J. London Math. Soc. (1982); **Schütte et al**, J. Comp. Phys. (1999)

Spectral Analysis

- **Eigenfunctions** encode metastable states:

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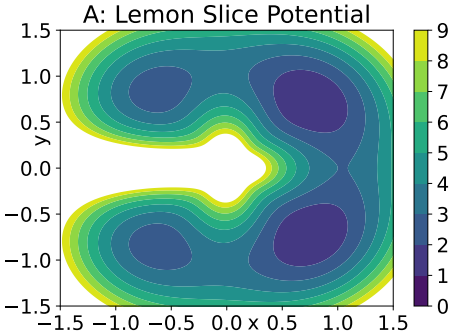

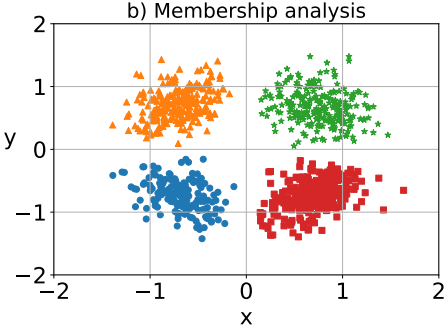

Davies, J. London Math. Soc. (1982); **Deuflhard & Weber**, Linear Algebra Appl. (2005)

Extended Dynamic Mode Decomposition (EDMD)

Track dynamics of **observables** by a **linear map**:

- Choose finitely many observables:

$$\psi(x) = \left[\psi_1(x) \cdots \psi_n(x)\right]^\top .$$

- Generate transformed snapshot matrices (x_k, y_k separated by **lag time** t):

$$\Psi(\mathbf{X}) = \left[\psi(x_1) \;\middle|\; \cdots \;\middle|\; \psi(x_m)\right], \quad \Psi(\mathbf{Y}) = \left[\psi(y_1) \;\middle|\; \cdots \;\middle|\; \psi(y_m)\right] \in \mathbb{R}^{n\times m}.$$

Williams et al, J. Nonlinear Science (2015);

Connection to VAC: **Klus, FN, Koltai, Wu, Kevrekidis, Schütte, Noé**, J. Nonlinear Sci. (2018).

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- Solve **regression** problem (EDMD):

$$\begin{aligned}\mathbf{K}^t &= \operatorname{argmin}_{\mathbf{K}\in\mathbb{R}^{n\times n}} \|\Psi(\mathbf{Y}) - \mathbf{K}^\top\Psi(\mathbf{X})\|_F \\ &= (\Psi(\mathbf{X})\Psi(\mathbf{X})^\top)^{-1}(\Psi(\mathbf{X})\Psi(\mathbf{Y})^\top).\end{aligned}$$

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Generator Learning (gEDMD)

- Koopman generator (overdamped Langevin):

$$\mathcal{L}\psi = -\nabla V \cdot \nabla\psi + k_B T \Delta\psi.$$

- Data-driven matrix model analogous to EDMD:

$$\Psi(\mathbf{X}) = \begin{bmatrix} \psi_1(x_1) & \dots & \psi_1(x_m) \\ \vdots & \ddots & \vdots \\ \psi_n(x_1) & \dots & \psi_n(x_m) \end{bmatrix}, \qquad \mathcal{L}\Psi(\mathbf{X}) = \begin{bmatrix} \mathcal{L}\psi_1(x_1) & \dots & \mathcal{L}\psi_1(x_m) \\ \vdots & \ddots & \vdots \\ \mathcal{L}\psi_n(x_1) & \dots & \mathcal{L}\psi_n(x_m) \end{bmatrix},$$

- matrix model:

$$\mathbf{L} = \left(\Psi(\mathbf{X})\Psi(\mathbf{X})^\top\right)^{-1}\left(\Psi(\mathbf{X})\mathcal{L}\Psi(\mathbf{X})^\top\right).$$

Klus, FN, Peitz, Niemann, Clementi, Schütte, Physica D, 2020.

Generator Learning (gEDMD)

- Data-requirements for gEDMD:

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- trajectory integration not required

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- trajectory integration not required
- Remark: mass matrix can be formed in first order form:

$$\left(\Psi(\mathbf{X})\mathcal{L}\Psi(\mathbf{X})^\top\right) = \frac{1}{2m}\sum_{k=1}^{m} \nabla\psi(x_k)\cdot\nabla\psi(x_k)^\top.$$

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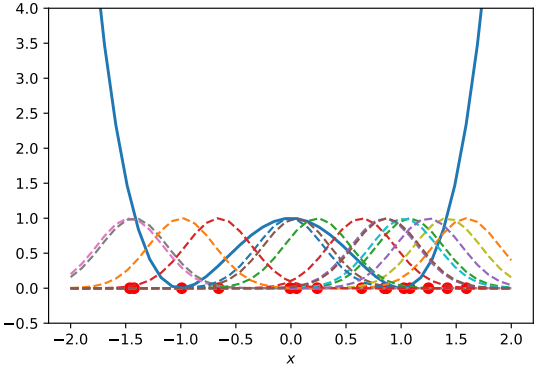
Basis Sets and Kernel

- Can we use a basis defined by the data?

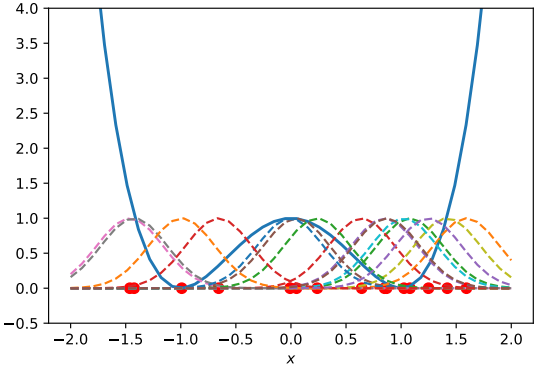


Berlinet, Thomas-Agnon (2004); **Wendland** (2005); **Steinwart, Christmann** (2008);...

Basis Sets and Kernel

- Can we use a basis defined by the data?
- Typical choice: **radial basis functions**, e.g.

$$\psi_i(y) = k_\sigma(x_i, y) = \exp\left(-\frac{1}{2\sigma^2}\|x_i - y\|^2.\right)$$



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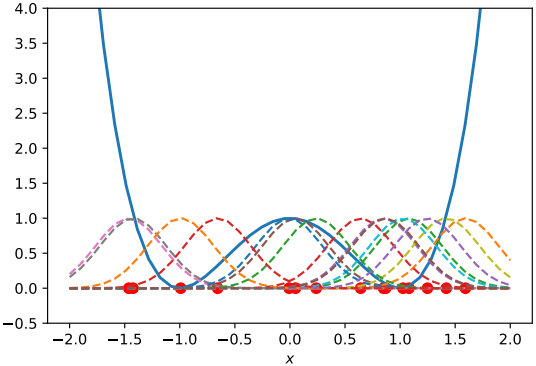
- Can use any **positive-definite kernel** $k(x, y)$ with

$$\psi_i(y) = k(x_i, y).$$



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EDMD and Kernels

- Using the basis set $\psi_i = k(x_i, \cdot)$, for $1 \leq i \leq m$, and plugging into the EDMD equations, we obtain the matrix representation:

$$\mathbf{K}_{\mathbb{H}}^t = \mathbf{K}_{\mathrm{XX}}^{-1}\mathbf{K}_{\mathrm{YX}}, \qquad \mathbf{K}_{\mathrm{XX}}(r,s) = k(x_r, x_s), \qquad \mathbf{K}_{\mathrm{YX}}(r,s) = k(y_r, x_s).$$

Klus, Schuster, Muandet, J. Nonlinear Science (2020), **FN, Klus, Hamzi**, Entropy (2020)

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- For gEDMD, replace (reversible) mass matrix by

$$\mathbf{K}_{\mathrm{d}X}(r,s) = \frac{1}{2m}\sum_{k=1}^{m} \nabla k(x_r, x_k) \cdot \nabla k(x_k, x_s).$$

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- All matrices are of size $m \times m$. Hyper-parameter tuning can get expensive.

Klus, Schuster, Muandet, J. Nonlinear Science (2020), **FN, Klus, Hamzi**, Entropy (2020)

Random Fourier Features

- For a **translation-invariant** kernels...

$$k(x, y) = \gamma(x - y).$$

Rahimi and Recht, Advances in Neural Information Processing Systems, 2007
Wendland, Scattered Data Approximation (2005)

Random Fourier Features

- For a **translation-invariant** kernels...

$$k(x, y) = \gamma(x - y).$$

- ...write is as an inverse Fourier transform:

$$k(x, y) = \gamma(x - y) = (2\pi)^{-d/2} \int \hat{\gamma}(\omega) e^{i\omega^\top (x-y)} \, \mathrm{d}\omega.$$

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- We have $\hat{\gamma}(\omega) \geq 0$. If $\gamma(0) = k(x, x) = 1$, then $\mathrm{d}\rho = (2\pi)^{-d/2} \hat{\gamma}(\omega) \, \mathrm{d}\omega$ is a probability measure:

$$k(x, y) = \gamma(x - y) = \mathbb{E}^{\omega \sim \rho} \left[e^{i\omega^\top x} \left(e^{i\omega^\top y} \right)^* \right] \approx \frac{1}{p} \sum_{\ell=1}^{p} e^{i\omega_\ell{}^\top x} \left(e^{i\omega_\ell{}^\top y} \right)^*$$

- General statement is **Bochner's Theorem**.

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Wendland, Scattered Data Approximation (2005)

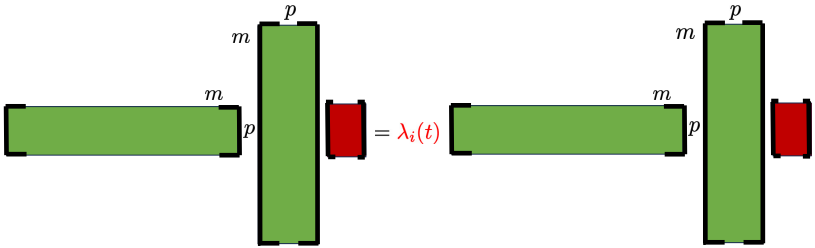
Dual Formulation

- Plug into kernel EDMD: (g)EDMD with **random plane wave basis**:

$$\Psi_\omega(x) = \begin{bmatrix} e^{i\omega_1^\top x} & \dots & e^{i\omega_p^\top x} \end{bmatrix} \in \mathbb{C}^p.$$

- Reduces to matrix of dimension p:

$$\mathbf{L} = \left(\Psi_\omega(\mathbf{X})\Psi_\omega(\mathbf{X})^\top\right)^{-1}\left(\Psi_\omega(\mathbf{X})\mathcal{L}\Psi_\omega(\mathbf{X})^\top\right)$$



FN and Klus, J. Chem. Phys., 2023

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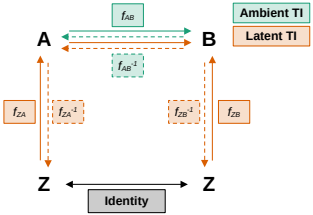
Thermodynamic Interpolation

- **Temperature-dependent** generative model trained on high-temperature data.
- Extrapolate to target temperature.

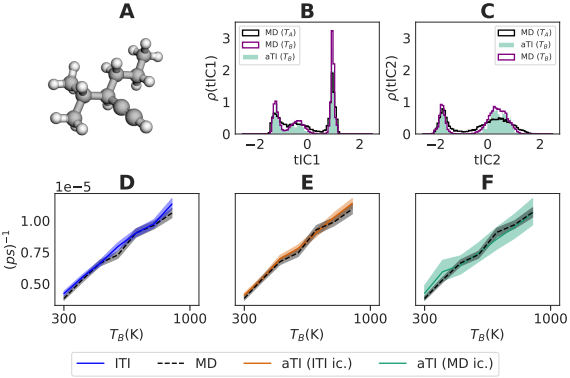


Noé, Olsson, Köhler, Wu, Science 2019,

Moqvist, Chen, Schreiner, FN and Olsson, J. Chem. Theory Comput., 2025

Thermodynamic Interpolation

- **Temperature-dependent** generative model trained on high-temperature data.
- Extrapolate to target temperature.
- Learn gEDMD model with random Fourier features on torsion angles.
- Extract **temperature dependent transition rates**.



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Moqvist, Chen, Schreiner, FN and Olsson, J. Chem. Theory Comput., 2025

Coarse Grained Generator

- Consider a **coarse-graining map** $\xi : \mathbb{R}^d \mapsto \mathbb{R}^n$, $n \leq d$.

Legoll and Lelièvre, Nonlinearity (2010),
Zhang, Hartmann, Schütte, Faraday Disc. (2016)

Coarse Grained Generator

- Consider a **coarse-graining map** $\xi : \mathbb{R}^d \mapsto \mathbb{R}^n$, $n \leq d$.
- Projected generator $\mathcal{L}^\xi = \mathcal{P}\mathcal{L}\mathcal{P}$ generates an **effective SDE**:

$$\mathrm{d}\mathcal{Z}_t = b^\xi(\mathcal{Z}_t)\,\mathrm{d}t + \sigma^\xi(\mathcal{Z}_t)\,\mathrm{d}W_t.$$

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- Fundamental identity:

$$\left\langle \psi_i \circ \xi,\, \mathcal{L}(\psi_j \circ \xi)\, e^{-\beta V} \right\rangle = \left\langle \psi_i,\, \mathcal{L}^\xi \psi_j\, e^{-\beta F} \right\rangle.$$

Legoll and Lelièvre, Nonlinearity (2010),
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Learning CG Dynamics

- CG dynamics in **reversible** form:

$$\mathrm{d}\mathcal{Z}_t = \left[-\frac{1}{2}a^{\xi}(\mathcal{Z}_t)\nabla F(\mathcal{Z}_t) + \frac{1}{2}\nabla\cdot a^{\xi}(\mathcal{Z}_t)\right]\mathrm{d}t + \sigma^{\xi}(\mathcal{Z}_t)\,\mathrm{d}W_t.$$

- Effective diffusion is learned by **diffusion matching**:

$$a^{\xi} = \arg\min_{a^{\theta}} \frac{1}{m}\sum_{i=1}^{m}\left\|a^{\theta}(\xi(x_i)) - a^{\xi}_{\mathrm{loc}}(x_i)\right\|_F^2,$$

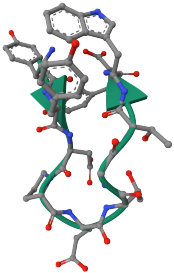
- gEDMD model for CG dynamics can be directly computed:

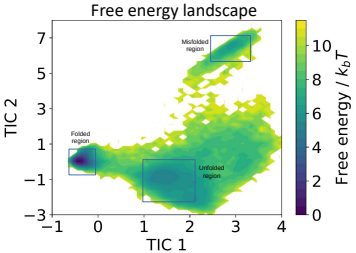
$$\left\langle \psi_r,\, \mathcal{L}^{\theta}\psi_s\, e^{-\beta F}\right\rangle_{\nu} = -\frac{1}{2}\int \nabla\psi_r(z)a^{\theta}(z)\nabla\psi_s(z)^{\top}\, e^{-\beta F(z)}\,\mathrm{d}z.$$

Nateghi and FN, *arxiv 2409.16396* (2024)

Learning CG Dynamics

- Coarse-graining of Chignolin mini-protein.
- Project into 2d TICA space; minimal effect of Markovian approx.





Charron et al, arXiv:2310.18278 (2023)
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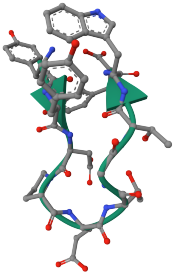
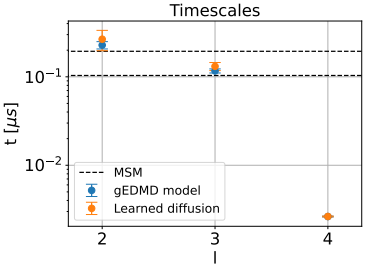
- Compute transition timescales of CG dyn.





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Learning CG Dynamics

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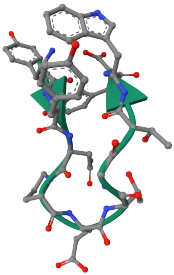
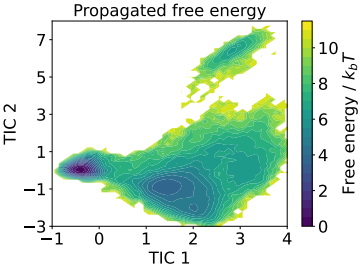
- Project into 2d TICA space; minimal effect of Markovian approx.
- Learn effective diffusion with RFFs.
- Compute transition timescales of CG dyn.
- Run simulations in TICA space; recover free energy.





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Thank you!

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